Relative Classical Cosmology

A metaphysical investigation of nonlocal nonalgorithmic dynamics

Draft 1.1

PREFACE

Albeit It does move.
[*apparently muttered as an act of defiance against torturers abhorrent towards such proponents of heliocentrism*]

Galileo Galilei

This interconnection or accommodation of all created things to each other, and each to all the others, brings it about that each simple substance has relations that express all the others, and consequently, that each simple substance is a perpetual, living mirror of the universe.

Gottfried Wilhelm Leibniz

Nature exists only once. It is only our schematizing reproduction that creates equal cases. It is only in those that there exists mutual dependence of certain qualities.

Ernst Mach

Mathematics is the art of giving the same name to different things.

Henri Poincaré

My [algebraic] methods are really methods of working and thinking; this is why they have crept in everywhere anonymously.

Emmy Noether

... of the actual objects of physical reality. Physical objects are not in space, but these objects are spatially extended.

Albert Einstein

I regard space as a 'glue', or a set of rules, that binds things together. It is a plurality within a deep unity, and it makes a Now.

Julian Barbour

Space and time emerge from the laws rather than providing an arena in which things happen.

Lee Smolin

Once again, the world seems to be less about objects than about interactive relationships.

Carlo Rovelli

... the relation of mathematics to the world of temporal change and of phenomenal particularity is direct: less by induction than by what Pierce called abduction – an imaginative jumping off from an open-ended series of particulars.

Roberto Mangabeira Unger

This preface begins with an unusual number of quotes. But what is 'number'? If numbers literally *enumerate* distinct, somewhat stable objects or processes in our thoughts and potentially their worldly referents, then it turns out that there is (somewhat) only *one* major message to be gleaned above, at least for the purposes of this book. And this book is about the story of that message, encoded in various representations, from mathematical to philosophical (there is nothing fancy in this combination; I simply happen to be an embodied spirit in a finite body, unable to access representation-free nature!).

But, we must not get ahead of ourselves, and we certainly must not interpret the above statement in any Pythagorean, let alone overly structural sense. It is simply a metaphor for a common story, a patchwork of clashing narratives, that stretches far out into the annals of the history of natural philosophy itself — even beyond the point where the historical character of philosophy and the philosophical character of history blur into many a psychophysical mess.

And, I somehow find myself compelled to dirty my hands at an unmovable mote-of-a-dust task lying in this constellation of the unknown unknown. And the unmovable mote of a task is to unfold a story-within-a-story-within-...-within-the-story.[1] But, I will try to "and-yet-it-moves" my way out of this.

If there's ever a way out.

And that's not necessarily a bad thing. To be part of nature, to be aware of at least some parts of it, is an experience which always seems to evade structure. And yet, this temporal, phenomenally soaked process of Cartesian knowing is one of the few ways we know the universe can know itself. So, maybe Platonic knowledge from 'outside' the temporal world, useful as it is as a heuristic, only applies to 'small' enough subsystems of the universe. Maybe mathematical, certain knowledge is only the beginning of a different kind of knowledge. That knowledge is perhaps that of Time.

[1] It amuses me that if nature is ultimately teleological, this Russian doll of a story might start looking like a plan-within-a-plan-within-a-..., such as the ones so curiously employed in Frank Herbert's *Dune*.

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CHAPTER 1

ORIENTED HYPERGRAPHS

1 Basic structure

Let X be a set. X is an **oriented hypergraph** if it is equipped with *oriented hyperedges*, i.e. a collection of (inherently oriented) tuples of *subsets* of X,

Definition 1.1: Oriented hypergraph

$(X, \mathcal{E})$ is an **oriented hypergraph** if,

$$\mathcal{E} \subseteq \bigcup_{k \in \mathbb{N}} \left(\mathcal{P}\left(X\right)\right)^k$$

Definition 1.2: Oriented k-hyperedges

Given an oriented hypergraph $(X, \mathcal{E})$, its oriented **k-hyperedges** or hyperedges of **k-arity** are elements of the disjoint $\mathbb{N}$-indexed sets,

$$\forall\, k \in \mathbb{N} : \mathcal{E}_k = \left(\mathcal{P}\left(X\right)\right)^k \cap \mathcal{E}$$

where $\mathcal{E}_0 = \emptyset$.

Instead of defining order and degree for an entire oriented hypergraph, it will be useful to define it for 'oriented sub-hypergraphs' of sorts, of the original graph. However, since we are trying to construct order-theoretic notions (like posets produced by structure inclusion) *from* oriented hypergraphs in the first place, we will not explicitly define oriented sub-hypergraphs but simply look at subsets of the set of oriented hyperedges of an oriented hypergraph.

Definition 1.3: Degree of subsets of oriented hyperedges

We define the **degree** of a subset of oriented hyperedges as its greatest arity, i.e. for all $E \subseteq \mathcal{E}$,

$$\deg(E) = \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}\left(X\right)\right)^k \cap E \neq \emptyset\right\}\right)$$

Definition 1.4: Extension and order of subsets of oriented hyperedges

The **extension** and **order** of a subset $E \subseteq \mathcal{E}$ are the set of elements of X referred to in elements of E, and its cardinality, respectively, i.e. for all $E \subseteq \mathcal{E}$,

$$\begin{aligned}
\operatorname{ext}(E) &= \left\{x \in X \middle| \begin{array}{l} \exists\, i < k \leq \deg(E) : \\ \end{array} \begin{array}{c} \exists\, A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k \subseteq X \\ : x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in E \end{array}\right\} \\
\operatorname{ord}(E) &= |\operatorname{ext}(E)|
\end{aligned}$$

One can think of the above as classifications of regions of hypergraphs, and classifications of new regions constructed from older ones (via set-theoretic operations) can be found algebraically.

Theorem 1.5: Degree, extension and order under hyperedge subset unions and intersections

For all $A, B \subseteq \mathcal{E}$,

1. $\deg(A \cup B) = \max\left(\{\deg(A), \deg(B)\}\right)$
2. $\deg(A \cap B) \leq \min\left(\{\deg(A), \deg(B)\}\right)$
3. $\operatorname{ext}(A \cup B) = \operatorname{ext}(A) \cup \operatorname{ext}(B)$
4. $\operatorname{ext}(A \cap B) = \operatorname{ext}(A) \cap \operatorname{ext}(B)$
5. $\operatorname{ord}(A \cup B) = \operatorname{ord}(A) + \operatorname{ord}(B) - \operatorname{ord}(A \cap B)$

Proof. We have:

1.

$$\begin{aligned}
\deg(A \cup B) &= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap (A \cup B) \neq \emptyset\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\left(\mathcal{P}(X)\right)^k \cap A\right) \cup \left(\left(\mathcal{P}(X)\right)^k \cap B\right) \neq \emptyset\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right) \vee \left(\left(\mathcal{P}(X)\right)^k \cap B \neq \emptyset\right)\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right\} \cup \left\{l \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^l \cap B \neq \emptyset\right\}\right) \\
&= \max\left(\left\{\max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right\}\right), \max\left(\left\{l \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^l \cap B \neq \emptyset\right\}\right)\right\}\right) \\
&= \max\left(\{\deg(A), \deg(B)\}\right)
\end{aligned}$$

2. Consider the following two exhaustive cases (i.e., at least one of them must be true):

   - $A \cap B \neq \emptyset$. Then,

$$\begin{aligned}
\deg(A \cap B) &= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \cap B \neq \emptyset\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right) \wedge \left(\left(\mathcal{P}(X)\right)^k \cap B \neq \emptyset\right) \wedge (A \cap B \neq \emptyset)\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right) \wedge \left(\left(\mathcal{P}(X)\right)^k \cap B \neq \emptyset\right)\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right\} \cap \left\{l \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^l \cap B \neq \emptyset\right\}\right)
\end{aligned}$$

   Since the two sets (which we intersect and take the maximum of) above intersect, the maximum arity of their intersection cannot exceed the maximum arity of the set with the lower degree. Hence,

$$\begin{aligned}
\deg(A \cap B) &\leq \min\left(\left\{\begin{array}{c}\max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \neq \emptyset\right\}\right), \\ \max\left(\left\{l \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^l \cap B \neq \emptyset\right\}\right)\end{array}\right\}\right) \\
&\leq \min\left(\{\deg(A), \deg(B)\}\right)
\end{aligned}$$

   - $A \cap B = \emptyset$. Then,

$$\begin{aligned}
\deg(A \cap B) &= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap A \cap B \neq \emptyset\right\}\right) \\
&= \max\left(\left\{k \in \mathbb{N} \middle| \left(\mathcal{P}(X)\right)^k \cap \emptyset \neq \emptyset\right\}\right) \\
&= \max\left(\emptyset\right) \\
&:= 0 \\
&\leq \min\left(\{\deg(A), \deg(B)\}\right)
\end{aligned}$$

3. We have,

$$
\begin{aligned}
\operatorname{ext}(A \cup B) &= \left\{ x \in X \,\middle|\, \begin{array}{l} \exists\, i < k \leq \deg(E) : \\ \end{array} \begin{array}{c} \exists\, A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k \subseteq X \\ : x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in A \cup B \end{array} \right\} \\
&= \left\{ x \in X \,\middle|\, \begin{array}{l} \exists\, i < k \leq \deg(E) : \\ \end{array} \begin{array}{c} \exists\, A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k \subseteq X \\ : (x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in A) \vee \\ (x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in B) \end{array} \right\} \\
&= \left\{ x \in X \,\middle|\, \begin{array}{l} \exists\, i < k \leq \deg(E) : \\ \end{array} \begin{array}{c} \exists\, A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k \subseteq X \\ : x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in A \end{array} \right\} \\
&\cup \left\{ x \in X \,\middle|\, \begin{array}{l} \exists\, i < k \leq \deg(E) : \\ \end{array} \begin{array}{c} \exists\, A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k \subseteq X \\ : x \in U, (A_1, \ldots, A_{i-1}, U, A_{i+1}, \ldots, A_k) \in B \end{array} \right\} \\
&= \operatorname{ext}(A) \cup \operatorname{ext}(B)
\end{aligned}
$$

4. By similar reasoning as above, we can write out $\operatorname{ext}(A \cap B)$ explicitly, decompose the quantification of the set into a *conjunction* (instead of a disjunction) and take the union of the individual sets to find:

$$\operatorname{ext}(A \cap B) = \operatorname{ext}(A) \cap \operatorname{ext}(B)$$

5. From the above, it follows that for disjoint subsets of oriented hyperedges, $A \cap B = \emptyset$, therefore $\operatorname{ext}(A \cap B) = \operatorname{ext}(\emptyset) = \emptyset$, so we must also have disjoint extensions, i.e. $\operatorname{ext}(A) \cap \operatorname{ext}(B) = \emptyset$. With that in mind,

$$
\begin{aligned}
\operatorname{ord}(A \cup B) &= |\operatorname{ext}(A \cup B)| \\
&= |\operatorname{ext}(A \cup B \backslash A)| \\
&= |\operatorname{ext}(A) \cup \operatorname{ext}(B \backslash A)| \\
&= |\operatorname{ext}(A)| + |\operatorname{ext}(B \backslash A)| && [A \cap B \backslash A = \emptyset] \\
&= \operatorname{ord}(A) + \operatorname{ord}(B \backslash A) \\
\text{But } \operatorname{ord}(B) &= |\operatorname{ext}(B)| \\
&= |\operatorname{ext}((B \backslash A) \cup (A \cap B))| \\
&= |\operatorname{ext}(B \backslash A) \cup \operatorname{ext}(A \cap B)| \\
&= |\operatorname{ext}(B \backslash A)| + |\operatorname{ext}(A \cap B)| && [(B \backslash A) \cap (A \cap B) = \emptyset] \\
&= \operatorname{ord}(B \backslash A) + \operatorname{ord}(A \cap B) \\
\implies \operatorname{ord}(B \backslash A) &= \operatorname{ord}(B) - \operatorname{ord}(A \cap B) \\
\operatorname{ord}(A \cup B) &= \operatorname{ord}(A) + \operatorname{ord}(B) - \operatorname{ord}(A \cap B)
\end{aligned}
$$

□

Corollary 1.6

For any $A \subseteq B \subseteq \mathcal{E}$, we have:

1. $\deg(A) \leq \deg(B)$
2. $\text{ext}\,(A) \subseteq \text{ext}\,(B)$
3. $\text{ord}\,(A) \leq \text{ord}\,(B)$

Proof. Consider:

1.

$$\begin{aligned}\deg(A) &= \deg(A \cap B) \\ &\leq \min\left(\{\deg(A), \deg(B)\}\right) \\ &\leq \deg(B)\end{aligned}$$

2.

$$\begin{aligned}\text{ext}\,(A) &= \text{ext}\,(A \cap B) \\ &= \text{ext}\,(A) \cap \text{ext}\,(B) \\ &\subseteq \text{ext}\,(B)\end{aligned}$$

3. It follows from the above that $|\text{ext}\,(A)| \leq |\text{ext}\,(B)|$, which is identical to the statement $\text{ord}\,(A) \leq \text{ord}\,(B)$.

□

Theorem 1.7: Arities up to overall degree form a partition for set of oriented hyperedges

$\mathcal{E}$ is the union of oriented hyperedges over all arities up to its overall degree,

$$\mathcal{E} = \bigcup_{k \leq \deg(E)} \mathcal{E}_k$$

with $\mathcal{E}_k \cap \mathcal{E}_l = \emptyset$ for all $k \neq l$. In other words, $\{\mathcal{E}_k | k \leq \deg(\mathcal{E})\}$ is a partition for $\mathcal{E}$.

Proof.

$$\begin{aligned}\mathcal{E} &= \left(\bigcup_{k\in\mathbb{N}} (\mathcal{P}(X))^k\right) \cap \mathcal{E} \\ &= \bigcup_{k\in\mathbb{N}} \left[(\mathcal{P}(X))^k \cap \mathcal{E}\right] \\ &= \bigcup_{k\leq \deg(\mathcal{E})} \left[(\mathcal{P}(X))^k \cap \mathcal{E}\right] \\ &= \bigcup_{k\leq \deg(E)} \mathcal{E}_k\end{aligned}$$

Since for any $k \neq l$, we must have $(\mathcal{P}(X))^k \cap (\mathcal{P}(X))^l = \emptyset$, it follows that,

$$\begin{aligned}\mathcal{E}_k \cap \mathcal{E}_l &= \left((\mathcal{P}(X))^k \cap \mathcal{E}\right) \cap \left((\mathcal{P}(X))^l \cap \mathcal{E}\right) \\ &= (\mathcal{P}(X))^k \cap (\mathcal{P}(X))^l \cap \mathcal{E} \\ &= \emptyset \cap \mathcal{E} \\ &= \emptyset\end{aligned}$$

□

Corollary 1.8

1. $\operatorname{ext}(\mathcal{E}) = \bigcup_{k\leq \deg(\mathcal{E})} \operatorname{ext}(\mathcal{E}_k)$
2. $\operatorname{ord}(\mathcal{E}) = \sum_{k\leq \deg(\mathcal{E})} \operatorname{ord}(\mathcal{E}_k)$

Proof. Since $\{\mathcal{E}_k | k \leq \deg(\mathcal{E})\}$ is a partition for $\mathcal{E}$,

$$\cup \{\mathcal{E}_k | k \leq \deg(\mathcal{E})\} = \mathcal{E}$$

and,

$$\cup \{\operatorname{ext}(\mathcal{E}_k) | k \leq \deg(\mathcal{E})\} = \operatorname{ext}(\mathcal{E})$$

with the set in L.H.S. above still being a partition for that in R.H.S. Hence, when we take the cardinality of both sides and interchange L.H.S. and R.H.S., we recover:

$$\operatorname{ord}(\mathcal{E}) = \sum_{k\leq \deg(\mathcal{E})} \operatorname{ord}(\mathcal{E}_k)$$

□

Given oriented hyperedges of any arity, we can further classify them based on the cardinalities of subsets of X they relate:

Fact 1.9

For any $k \in \mathbb{N}$, an oriented hyperedge in $\mathcal{E}_k$ is of **rank** $(\tau_1, \ldots, \tau_k)$ where for all $i \leq k$,

$$\tau_i = \max\left(\{|A_i| : A_i \subseteq X, \exists\, A_1, \ldots, A_{i-1}, A_{i+1}, \ldots, A_k \subseteq X : (A_1, \ldots, A_i, \ldots, A_k) \in \mathcal{E}_k\}\right)$$

The rank of some $e \in \mathcal{E}_k$ is denoted as $\mathrm{rk}\left(\{e\}\right)$, with its i-th element being represented as $\mathrm{rk}_i\left(\{e\}\right)$.

Intuitively, an oriented k-hyperedge is a sequence of subsets of X, with the corresponding sequence of their maximum allowed cardinalities given by the rank. The above facts lead to the following abstract definition.

Definition 1.10: Formal definition of ranks of oriented hyperedges

$\mathrm{rk} : \mathcal{P}\left(\mathcal{E}\right) \to \bigcup_{k \in \mathbb{N}} \mathbb{N}^k$ is a function such that it satisfies:

1. *Trivial action on arity partitions*: $\forall\, k \in \mathbb{N} : \mathrm{rk}\left(\mathcal{E}_k\right) = k$
2. *Order preservation*[a]: $\forall\, E, E' \subseteq \mathcal{E} : E \subseteq E' \implies \forall\, i \leq \deg(E) : \mathrm{rk}_i\left(E\right) \leq \mathrm{rk}_i\left(E'\right)$

[a] From the perspective of order theory, one can say that rank is order-preserving in that the rank tuple for E is a sub-tuple of the rank tuple for E', *and* that for corresponding elements, the ones in E' are always greater in value; since both of these notions are order-preserving on tuples, so is their composition.

$(X, \mathcal{E})$ is a *directed hypergraph* if $\deg(\mathcal{E}) = 2$ and $\mathcal{E}_1 = \emptyset$, i.e. all hyperedges possess *direction*, i.e. the simplest possible notion of orientation, only by virtue of 2-ary relationships between subsets of X.

On the other hand, X is an *undirected hypergraph* if $\deg\left(X\right) = 1$, i.e. the hyperedges have no direction via 2-ary or orientation via higher-degree relationships between subsets of X. Equivalently, $\mathcal{E}$ is simply a collection of subsets of X.

X is a *graph* if it is a directed hypergraph of rank unity, $\mathrm{rk}\left(\mathcal{E}\right) = (1, 1)$, i.e. the edges simply model a binary relationship on X itself. Hence, a *directed graph* models an antisymmetric binary relationship, whereas an *undirected graph* models a symmetric binary relationship.

Unless explicitly mentioned otherwise, we will refer to directed hypergraphs as simply 'hypergraphs' and to directed graphs as 'graphs'.

2 Automatic labelling

If X is a hypergraph with hyperedges in $\mathcal{E}$, X is *labelled* if there is a set of *labels* L and *labelling functions* $\phi : \mathcal{E} \to L$.

Definition 2.1: Labelled hypergraphs

$(X, \mathcal{E}, L, \phi)$ is an L**-labelled hypergraph** if $(X, \mathcal{E})$ is a hypergraph, L is a set and $\phi : \mathcal{E} \to L$ is a function.

The multi-layered structure of labelled hypergraphs leads to some interesting theorems which helps represent different kinds of them in multiple ways.

Theorem 2.2: Hypergraphs correspond to truth-value-labelled maximal hypergraphs

Let $(X, \mathcal{E})$ be a hypergraph. The **maximal hypergraph** corresponding to it is $\left(X, \mathcal{E}_X^{(\max)}\right)$ where,

$$\mathcal{E}_X^{(\max)} = \bigcup_{k \in \mathbb{N}} \left(\mathcal{P}\left(X\right)\right)^k$$

Since the above exists, so does the labelled maximal hypergraph $\left(X, \mathcal{E}_X^{(\max)}, 2, \phi\right)$ where ϕ is given by the predicate (with 2 as its extension, i.e. the set of truth values $\{\top, \bot\}$):

$$\forall\, k \in \mathbb{N} : \forall\, A_1, \ldots, A_k \subseteq X : \phi\left(\left(A_1, \ldots, A_k\right)\right) : \iff \left(A_1, \ldots, A_k\right) \in \mathcal{E}_k$$

In other words, one can represent $(X, \mathcal{E})$ as a different type of object possessing the same structure, $\phi : \mathcal{E}_X^{(\max)} \to 2$.

Informally, the above theorem justifies the back-and-forth switching between explicitly using hyperedges and representing them via abstract functions, in propositions concerning only the extensional structure of a hypergraph.

Theorem 2.3: Multilabelled hypergraphs correspond to labelled product hypergraphs

Consider a ***I*-multilabelled hypergraph**, i.e. a structure $(X, \mathcal{E}, L_1, L_2, \dots, \phi_1, \phi_2, \dots)$ where for each $k \in I \subseteq \mathbb{N}$, we have that $(X, \mathcal{E}, L_k, \phi_k)$ is a labelled hypergraph. The above structure can be constructed using the information in the single *product* hypergraph:

$$\left(X, \mathcal{E}, \prod_{k \in I} L_k, \bigotimes_{k \in I} \phi_k\right)$$

where,

$$\forall\, k \in \mathbb{N} : \forall\, A_1, \dots, A_k \in \mathcal{E}_k : \left(\bigotimes_{j \in I} \phi_j\right)(A_1, \dots, A_k) = (\phi_i(A_1, \dots, A_k) | i \in I)$$

Informally, this theorem justifies the usage of labelled hypergraphs *simpliciter* even in contexts involving multiple labels at the bottom level.

The two above theorems lead to the following powerful result: **multi-hypergraphs correspond to labelled hypergraphs**,

Corollary 2.4

Consider a hypergraph $(X, \mathcal{E})$ where it is possible to have multiple copies of any given hyperedge with their own labels, i.e., the structure is that of a **labelled hypermultigraph**. Let us define the **degeneracy** of a hyperedge as the number of its copies:

$$D : \begin{cases} \mathcal{E} & \to \mathbb{N} \\ (A_1, \dots, A_k) & \mapsto D\left((A_1, \dots, A_k)\right) \end{cases}$$

where $A_1, \dots, A_k \subseteq X$. The i-th copy of each hyperedge A would possess a label in some label set $L_{A,i}$, so we want a labelling map ϕ like so:

$$\forall\, A \in \mathcal{E} : \forall\, 0 \leq i \leq D(A) : \phi(A, i) \in L_{A,i}$$

Instead of constructing the above structure as (multiple) labels on $(X, \mathcal{E}, \mathbb{N}, D)$, we can equivalently specify it as $\left(X, \bigcup_{A \in \mathcal{E}} \{\phi(A, i) | 0 \leq i \leq D(A)\}, \bigcup_{A \in \mathcal{E}} \{L_{A,i} | 0 \leq i \leq D(A)\}, \phi\right)$.

Intuitively, the above is a consequence of there being a bijection between $\mathbb{N}$ and $\{\,(k | i \leq k) | k \in \mathbb{N}\}$. Hence, instead of listing degenerate hyperedges one by one with their labels (corresponding to the latter set), one can label the equivalence class of degenerate hyperedges with the indices counting degeneracy paired up with labels, which corresponds to $\mathbb{N}$.

Finally, we have a super-corollary: **labelled hypergraphs correspond to product**

hypergraphs.

Corollary 2.5

If $(X, \mathcal{E}, L, \phi)$ is a labelled hypergraph, its structure corresponds to that of the *unlabelled* hypergraph $(X \times L, \psi \otimes \phi)$ where $(X, \mathcal{E})$ is modelled by $\psi : \mathcal{E}_X^{(\max)} \to 2$ as,

$$\forall\, k \in \mathbb{N} : \forall\, A_1, \ldots, A_k \subseteq X : (A_1, \ldots, A_k) \in \mathcal{E}_k \iff \psi((A_1, \ldots, A_k))$$

Note that as a further simplification, the truth-value-labelled maximal hypergraph corresponding to every hypergraph, also corresponds to a hypermultigraph where a degeneracy of 0 means that the corresponding hyperedge is not to be included in the actual hypergraph.

To summarize, **hypergraphs *simpliciter* contain enough relational structure to model multilabelled hypermultigraphs using functions!** This will be useful in the upcoming sections.

3 Visual representation

Let us now visualize and reinterpret the above results using tabular and diagrammatic representation.

A directed (unlabelled) hypergraph $(X, \mathcal{E})$ can be represented by explicitly laying out the elements of X and listing the subsets in each $\mathcal{E}_k$, such as below.

X	$\mathcal{E}_1$				$\mathcal{E}_2$			
1	•			•	•		•	
2		•		|		•	|	•
3			•	•		•	•	

Figure 1.1: A tabular representation of an oriented hypergraph $(\{1, 2, 3\}, \mathcal{E}_1 \cup \mathcal{E}_2,)$ of degree 2 and rank $(2, 1)$.

From the above, we can explicitly list out hyperedges of all arities:

$$\mathcal{E}_1 = \{\{1\}, \{2\}, \{3\}, \{1, 3\}\}$$
$$\mathcal{E}_2 = \{(\{1\}, \{2\}), (\{\{3\}, \{2\}\}), (\{1, 3\}, \{2\})\}$$

We can 'interpret' the above structure as follows. $\mathcal{E}_1$ contains all subsets of X which do not simultaneously include even and odd elements. $\mathcal{E}_2 \subsetneq \mathcal{E}_1 \times \mathcal{E}_1$ contains directed hyperedges (i.e. oriented 2-hyperedges) which map such subsets with odd elements to those with even elements, i.e., $(\mathcal{E}_1, \mathcal{E}_2)$ is itself a **bipartite (directed) graph**[1].

[1] i.e., the hyperedges between elements of $\mathcal{E}_1$ represent a 2-partition on it.

To summarize the 'important' abstract *properties* of the above hypergraph:

1. $\mathcal{E}_1 = \bigcup_{[k]\in\mathbb{Z}_2} \mathcal{P}\left([k] \cap X\right)$ where $0 \leq k < 2$.

2. For all $A \in \mathcal{E}_1$ of rank $(1, \cdot)$, there exists some map $\varphi_A : (2\mathbb{Z})^{2\mathbb{Z}+1}$ s.t. $(A, \varphi_A(A)) \in \mathcal{E}_2$.

3. For all $U \in \mathcal{E}_1$ of rank $(2, \cdot)$, there exist rank $(1, \cdot)$ undirected hyperedges (in this case, disjoint singletons) $A, B \in \mathcal{E}_1$ s.t. $A \cup B = U$ and there exist $\varphi_A, \varphi_B \in (2\mathbb{Z})^{2\mathbb{Z}+1}$ s.t. the functional action of $\mathcal{E}_2$ on elements of $\mathcal{P}\left([1] \cap U\right)$ is to 'glue together' the actions of φ_A on A and of φ_B on B and produce new elements in $\mathcal{P}\left([0] \cap U\right)$,

$$\forall\, U \in \mathcal{E}_1 : (U, \varphi_U(U)) \in \mathcal{E}_2$$

where,

$$\varphi_U(U) = \mathrm{im}_{\varphi_A}(A) \cup \mathrm{im}_{\varphi_B}(B)$$

4. The above constructions exhaust the elements of $\mathcal{E}_2$.

Proof. Set $\varphi_{\{1\}}(\{k\}) = \{2k+1\}$ and $\varphi_{\{3\}}(\{l\}) = \{2l-1\}$. We indeed have that,

$$\begin{aligned}
\left(\{1\}, \varphi_{\{1\}}(\{1\})\right) &\in \mathcal{E}_2 \\
\left(\{3\}, \varphi_{\{3\}}(\{3\})\right) &\in \mathcal{E}_2 \\
\left(\{\{1\}, \{3\}\}, \varphi_{\{\{1\},\{3\}\}}\left(\{\{1\}, \{3\}\}\right)\right) &\in \mathcal{E}_2
\end{aligned}$$

where,

$$\begin{aligned}
\varphi_{\{\{1\},\{3\}\}}\left(\{\{1\}, \{3\}\}\right) &= \mathrm{im}_{\varphi_{\{1\}}}(\{1\}) \cup \mathrm{im}_{\varphi_{\{3\}}}(\{3\}) \\
&= \{2\} \cup \{2\} \\
&= \{2\}
\end{aligned}$$

□

Visually, we can capture the above information as follows:

$$\begin{tikzcd}
\{1\} \arrow[dr, "\varphi_{\{1\}}"] & \\
\cup \arrow[r, "\varphi_{\{\{1\},\{3\}\}}"] & \{2\} \\
\{3\} \arrow[ur, "\varphi_{\{3\}}"'] &
\end{tikzcd}$$

Figure 1.2: An alternative visual representation of the structure of the previous tabulated hypergraph.

Fact 3.1

Note that if we include the element $\emptyset$ in $\mathcal{E}_1$, then $\mathcal{E}_1$ is closed under intersections. Then, the various maps over various elements of $\mathcal{E}_1$ used can be combined in structure-preserving manner to generate new maps from entirely odd elements to entirely even elements, producing elements of $\mathcal{E}_2$.
This combinatorial structure underlying special oriented hypergraphs will be the basis for abstract simplicial complexes.

Using the visual scheme, some of the previous theorems on oriented hypergraphs can be justified intuitively. For example, the table for every hypergraph is a 'sub-table' of sorts of the table corresponding to the maximal hypergraph; the truth-value label for a given entry models *whether the entry in the maximal table actually appears in the hypergraph's table*.

Claim

Theorem **??** is intuitively a hypergraph-theoretic analogue of a combination of the power set axiom (maximal hypergraphs are in fact hypergraphs, just as power are sets) and the axiom of choice (entries from general hypergraphs, and hence maximal hypergraphs, can be collected together into valid hypergraphs, just as elements from sets, such as a power set, can be collected together into a new set).

The parallelism is similar for labelled hypergraphs. A labelled hypergraph would involve entering a label into each cell in the table for the hypergraph. Multiple labels per cell (in multilabelled hypergraphs) can be then collected together in a single tuple, which can be used as a new label. This is the informal verification for theorem **??**.

Furthermore, one can create multiple copies of each cell and label each, thereby modelling (multi)labelled hypermultigraphs.

Lastly, since the labels for each hyperedge go into its corresponding cell, one can imagine the superstructure to be identical to that of an *unlabelled* graph (formally speaking), where the labels *are* elements of the new cells in a product hypergraph.

4 Functional aspects

A common theme of the manipulations performed in the previous sections is that hypergraphs are rich in information about functions, at various levels of *built-in* versus emphadditionally supplied structure.

For example, every hypergraph trivially models a truth-value-labelled function on its maximal hypergraph (with the function selecting the former as a subset of the latter maximal hypergraph). On the other hand, one can supply functions between the extensions of two hypergraphs and construct a corresponding 'functional hypergraph'. This means that the functional nature of a given hypergraph is contingent when considering the nature of its (potentially valid) action on *other* hypergraphs (rather than such trivial cases as itself or its own maximal hypergraph).

Let us now formalize the non-trivial functional aspects of hypergraphs, in the form of restrictions, isomorphisms, embeddings and so on.

Definition 4.1: Restrictions of oriented hypergraphs

Let $(X, \mathcal{E})$ be an oriented hypergraph. Then, for any subset $U \subseteq X$, the corresponding **restriction** of $\mathcal{E}$ is the subset of hyperedges whose extension is contained in U,

$$\mathcal{E}|_U = \left(\bigcup_{k \in \mathbb{N}} \left(\mathcal{P}\left(U\right) \right)^k \right) \cap \mathcal{E}$$

Proof. The theorem implicit in the above definition is that it *is*, in fact, a definition, i.e. the above characterization of restrictions is a tautology. Such is the case iff $\operatorname{ext}\left(\mathcal{E}|_U\right) \subseteq U$, which we will now prove,

$$\begin{aligned}
\operatorname{ext}\left(\mathcal{E}|_U\right) &= \operatorname{ext}\left(\left(\bigcup_{k \in \mathbb{N}} \left(\mathcal{P}\left(U\right) \right)^k \right) \cap \mathcal{E} \right) \\
&= \operatorname{ext}\left(\bigcup_{k \in \mathbb{N}} \left(\mathcal{P}\left(U\right) \right)^k \right) \cap \operatorname{ext}\left(\mathcal{E}\right) \\
&= U \cap X \\
&= U
\end{aligned}$$

□

Definition 4.2: Functions on oriented hypergraphs

Consider two oriented hypergraphs, $(X, \mathcal{E}_X)$ and $(Y, \mathcal{E}_Y)$. Then, a set-theoretic function $f : X \to Y$ corresponds to the hypergraph $(X \times Y, \phi)$ yielding the hypergraph *simpliciter* $(X \times Y, \mathcal{E}_f)$ via the truth-function $\phi \in 2^{X \times Y}$ where,

$$\forall\, x \in X\, \forall\, y \in Y : (x, y) \in \mathcal{E}_f \iff \phi(x, y) \iff (y = f(x))$$

This means that given a collection of hypergraphs, only *some* of them (functional hypergraphs) have an appropriate functional structure with respect to *some* of the hypergraphs (background hypergraphs), in the case of which we can combine the latter using the former. This will become important for group theory and category theory.

Definition 4.3: Functional supergraphs of a family of hypergraphs

Let A be a set indexing a family $\mathcal{F}$ of (oriented) hypergraphs,

$$\mathcal{F} = \{(X_\alpha, \mathcal{E}_\alpha | \alpha \in A\}$$

Let $\mathcal{F}^*$ be the set of all strings formed from the elements of $\mathcal{F}$,

$$\mathcal{F}^* = \bigcup_{n \in \mathbb{N}} \mathcal{F}^k$$

Informally, a functional supergraph is a collection of hypergraphs which itself has a hypergraphical (relational structure) by virtue of how its elements can act on each other under different, equally valid systems of interpreting such actions as functions.

This is a primitive notion of both groups and categories, since:

1. A group involves the composition and action of objects.
2. A category involves actions on objects which can themselves be relational entities on a lower class of objects.

5 Some theorems

Before we study special classes of oriented hypergraphs as constituting different subfields of abstract algebra (for example, group theory) and geometry (for example, topological spaces), it will be useful to construct some overarching theorems in the domain of general oriented hypergraphs, which can then be applied to specific subfields to obtain lemmas and theorems with fairly general hypergraphical structure. Subsequently, these can be connected to corresponding structures in other subfields using the underlying hypergraph machinery, possibly even giving rise to intersectional subfields (conceivably algebraic topology, for instance).

First, some definitions:

6 Lattices

7 Combinatorial complexes

8 Groups

9 Topological spaces

10 Measure spaces

11 Dynamical systems

CHAPTER 2

ONTOLOGICAL GROUNDING

1 Ontologies

2 Mereology

3 Identity theory

4 Modality

5 Causation

6 Algorithm theory

7 Causal games

CHAPTER 3

LINEAR STRUCTURES ON BUNDLES

1 Introduction

1.1 Definition

Let $(B, \mathcal{O}_B)$ and $(E, \mathcal{O}_E)$ be topological spaces with a continuous projection[1] $\pi : E \to B$. By definition, (E, π, B) is a topological bundle.

Definition 1.1: Linear structures on bundles

A subset $L \subseteq E$ is a **linear structure** on (E, π, B) iff there exists a continuous curve $\gamma : B \to E$ such that L is the codomain of γ,

$$\exists\, \gamma \in \mathcal{C}(B, E) : \mathrm{im}_\gamma\,(B) = L$$

where we adopt the notation $\mathcal{C}^k(X, Y)$ for the class of functions $Y^X \cap C^k(X)$.

1.2 Induced topologies

Given such a continuous curve $\gamma : B \to E$ which indeed has L as its codomain, one can equip $L \subseteq E$ with the subset topology as follows,

$$\begin{aligned} \mathcal{O}_{L;\gamma} &:= \mathcal{O}_E|_L \\ &:= \{L \cap V | V \in \mathcal{O}_E\} \end{aligned}$$

Since γ is continuous, by definition, every open set $V \subseteq E$ has an open preimage in B under γ. The image of this preimage must return V, since γ is not merely a relation but also a *function*,

$$\begin{aligned} \mathcal{O}_{L;\gamma} &= \left\{\mathrm{im}_\gamma\,(B) \cap \mathrm{im}_\gamma\left(\mathrm{preim}_\gamma\,(V)\right) \middle| V \in \mathcal{O}_E\right\} \\ &= \left\{\mathrm{im}_\gamma\left(B \cap \mathrm{preim}_\gamma\,(V)\right) \middle| V \in \mathcal{O}_E\right\} \end{aligned}$$

[1] i.e., $\pi \circ \pi = \pi$.

Now, we ask, which open sets $\text{preim}_\gamma(V) \in \mathcal{O}_B$ are we quantifying over in the above set? I.e., which instances of these open sets in B have open images in the range? Namely, it is those open sets $U \in \mathcal{O}_B$ which are also preimages of E under γ, i.e. B itself. Consequently,

$$\begin{aligned}\mathcal{O}_{L;\gamma} &= \{\text{im}_\gamma(B \cap U) | U \in \mathcal{O}_B \cap B\} \\ &= \{\text{im}_\gamma(B \cap U) | U \in \mathcal{O}_B\} \\ &= \{\text{im}_\gamma(U) | U \in \mathcal{O}_B\}\end{aligned}$$

Theorem 1.2: Topologies induced on linear structures by special curves

If $L \subseteq E$ is a linear structure on a bundle (E, π, B), then every continuous curve $\gamma : B \to E$ satisfying $\text{im}_\gamma(B) = L$ endows L with the topology,

$$\mathcal{O}_{L;\gamma} = \{\text{im}_\gamma(U) | U \in \mathcal{O}_B\}$$

We immediately find rich application of the above concept stemming from a seemingly unstructured situation.

2 An example: the open unit interval

2.1 As a linear structure over $\mathbb{R} \xrightarrow{\pi} \mathbb{R}$ bundles

Consider the interval $(0, 1)$ and an arbitrary bundle $(\mathbb{R}, \pi, \mathbb{R})$.

Next, consider the class of functions $\Phi = [\mathcal{C}(\mathbb{R}, \mathbb{R}) \cap \mathcal{C}^0(\mathbb{R}, (0, \infty))] \cup \{0\}^{\mathbb{R}}$, i.e. continuous, strictly positive unbounded functions on $\mathbb{R}$ (plus the trivial null function[2]), exemplified by the decaying exponential $f(t) = e^{-t}$. By construction, we obtain a Φ-indexed family of continuous functions satisfying,

$$\forall f \in \Phi : \frac{1}{1+f} : \begin{cases} \mathbb{R} & \to (0,1) \\ t & \mapsto \frac{1}{1+f(t)} \end{cases}$$

Theorem 2.1: Φ indexes a family of topologies on $(0, 1)$

Each $f \in \Phi$ corresponds to a topology on $(0, 1)$,

$$\mathcal{O}_{(0,1);f} = \left\{ \left\{ \frac{1}{1+f(t)} \middle| t \in U \right\} \middle| U \in \mathcal{O}_{\mathbb{R}} \right\}$$

[2]Denoting this as 0_Φ gives $\{0\}^{\mathbb{R}} = \{0_\Phi\}$.

2.2 As producing a commutative division $\mathbb{R}$-algebra

One deep feature of the above sort of functional indexing is that we can upgrade the index set Φ to a *space* with additional structures satisfying useful properties like closure. This is an example of functional analysis.

Theorem 2.2: Φ is a vector space over $\mathbb{R}$

Let us upgrade Φ using functional addition $+ : \Phi^2 \to \Phi$ and *scalar multiplication* $\cdot : \mathbb{R} \times \Phi \to \Phi$. Quantifying over $s, t \in \mathbb{R}, f, g \in \Phi$,

$$(f+g)(t) := f(t) + g(t)$$
$$(s \cdot f)(t) := s[f(t)]$$

Then, $(\Phi, +, \cdot, \mathbb{R})$ is a vector space with $0_\Phi(t) = 0$ for all $t \in \mathbb{R}$.

Proof. It can be demonstrated that:

1. $\mathbb{R}$ is a commutative ring.
2. $(\Phi, +)$ is an abelian group with identity element 0_Φ.

   Quantifying over $r, s \in \mathbb{R}, f, g \in \Phi$, the following 'nice' distributive properties hold:
3. $r \cdot (f+g) = r \cdot f + r \cdot g$
4. $(r +_\mathbb{R} s)f = r \cdot f + s \cdot f$
5. $(rs) \cdot f = r \cdot (s \cdot f)$
6. $0_\Phi + f = f + 0_\Phi = f$

Hence, $(\Phi, +, \cdot, \mathbb{R})$ is an $\mathbb{R}$-module, but since $\mathbb{R}$ is a field, the structure narrows down to that of a vector space. □

The richness in the structure of Φ hardly stops with the above, as we also have the following!

Theorem 2.3: Φ is a commutative division $\mathbb{R}$-algebra

Define functional multiplication in Φ as follows,

$$\forall\, f, g \in \Phi : \forall\, t \in \mathbb{R} : (fg)(t) = f(t)g(t)$$

Then, Φ is a commutative division $\mathbb{R}$-algebra with the unital element $1_\Phi(t) = 1$ for all $t \in$R.

Proof. We have,

1. $\mathbb{R}$ is a commutative ring.
2. Φ is an $\mathbb{R}$-module.
3. $\mathbb{R}$-multiplication is compatible with functional multiplication, i.e., for all $r \in \mathbb{R}$ and $f, g \in \Phi$,
$$r \cdot (fg) = (r \cdot f)g = f(r \cdot g)$$
4. $1_\Phi f = f$
5. For all $f \neq 0_\Phi$, $\frac{1}{f} f = 1_\Phi$ where $\frac{1}{f}$ is the inverse element corresponding to f, given by[3],
$$\forall\, t \in \mathbb{R} : \left(\frac{1}{f}\right)(t) = \frac{1}{f(t)}$$

□

2.3 As producing a homeomorphism class on itself

A striking topological application of the above features of Φ is the following.

Theorem 2.4: Φ indexes a single homeomorphism class

All Φ-indexed topological spaces (as defined in theorem **??**) form a single homeomorphism class with respect to $(0, 1)$.

Proof. We must show that for any $f, g \in \Phi$, there exists a continuous bijection $h : (0, 1) \to (0, 1)$ s.t. h^{-1} is also continuous. Instead of doing this by brute force, functional analysis allows one to elegantly use the machinery of the functional indexing developed previously.

Specifically, we want to *algebraically* combine $f, g \in \Phi$ into $h\{f, g\} : (0, 1) \to (0, 1)$ in a way that satisfies closure[4], i.e. $h \in \Phi$. Since Φ supports division, Φ would then automatically include $\dfrac{1}{h}$. Then, if we can show that this algebraic inverse is also the *functional* inverse h^{-1}, the continuity of elements in Φ carries over to h^{-1}.

Let us construct such a scenario by allowing it to unfold 'backwards' (not logically!). The above scenario is logically compatible only with the following logical co-scenario:

$$\frac{1}{h} = h^{-1}$$
$$hh^{-1} = h^{-1}h = 1_\Phi$$

[3]Pedantically, the inverse of $f \in \Phi$ in a division algebra should be written as $\frac{1_\Phi}{f}$ to avoid abuse of notation, but this makes no practical difference with careful use, as one can always multiply by the scalar $1 \in \mathbb{R}$ in an associative algebra.

[4]Another deep structure in Φ is that $h\{f, g\}$ corresponds to a homotopy $\widetilde{h}[0, 1] \times \mathbb{R} \to \Phi$ continuously connecting f to g as a higher-order curve in Φ, i.e., $h(0, t) = f(t)$ and $h(1, t) = g(t)$. This is a link to homotopy theory.

This means that h, if it exists, must be in the centre $Z(\Phi) := \{m \in \Phi | \forall\, n \in \Phi : mn = nm\}$, which in this case is Φ itself, as it is a commutative ring. Furthermore, the choice for h, say an $\widetilde{h}$, does not matter as one can always define $h\{f, g\} = \widetilde{h}$. Hence, the algebraic inverse $\frac{1}{h}$ exists and is continuous, and since it was specially chosen to equal the functional inverse h^{-1}, the latter is continuous![5] □

2.4 As producing a class of ring homomorphisms $\mathbb{R} \to \Phi$ represented by invertible linear functionals $\Phi \to \mathbb{R}$

While the existence of trivial choices for $h\{f, g\}$, such as 1_Φ, complete the above proof, nontrivial properties of Φ can help construct very specific h-functions where f and g can appear explicitly. The task is then to find 'interesting' operators $h : \Phi^2 \to Z(\Phi) = \Phi$. Once again, the algebraic structure of Φ provides such an operator to us with a little setting up.

Recall that Φ is a commutative algebra. Hence, it comes with a ring homomorphism[6] $\varphi : \mathbb{R} \to \Phi$, for instance $\varphi(t) = t \cdot 1_\Phi$. We can then use the machinery of direct products to find the correct type of function corresponding to h. However, instead of restricting ourselves to the above characterization of φ, one can construct more general homomorphisms using linear functionals. Suppose L is an invertible linear functional $\Phi \to \mathbb{R}$ with a trivial kernel,

1. $\forall\, f, g \in \Phi : L[f + g] = L[f] + L[g]$
2. $\forall\, r \in \mathbb{R}\, \forall\, f \in \Phi : L[r \cdot f] = rL[f]$
3. $\forall\, r \in \mathbb{R} : \exists\, f \in \Phi : L[f] = r$
4. $\ker(L) = \{0_\Phi\}$

Firstly, consider some useful results from linear algebra:

Proposition 2.5

L is onto.

Proof. By construction, L is invertible. Therefore, L is onto. □

Proposition 2.6

L is one-to-one iff $\ker(L) = \{0_\Phi\}$.

[5] There is no logical circularity here as the reasoning works regardless of the choice of h from the centre $Z(\Phi) = \Phi$, and the former cannot be from any proper superset of the latter. In other words, the explanatory mode here takes the form wherein h has a nontrivial property that follows from nontrivial properties of a nontrivial equivalence class (here, the centre $Z(\Phi)$) containing h.

[6] which produces 'scalar functions' lying in the centre $Z(\Phi)$, which in this case is Φ itself.

Proof. Suppose $f, g \in \Phi$ map under L to the same $r \in \mathbb{R}$. Then,

$$\begin{aligned} L[f] = L[g] &= r \\ L[f] - L[g] &= 0 \\ L[f-g] &= 0 \\ f - g &\in \ker(L) \end{aligned}$$

But, if L is to be one-to-one, we cannot have $f \neq g$, which means $\ker(L)$ would have to be trivial.

On the other hand, if $\ker(L)$ is trivial but $f \neq g$ with $L[f] = L[g]$, then,

$$\begin{aligned} f - g &\neq 0_\Phi \\ L[f-g] &= 0 \\ \implies f - g &\in \ker(L) \\ \implies \ker(L) &\supsetneq \{0_\Phi\} \end{aligned}$$

which is a contradiction. Therefore, we must have $f = g$. □

But recall that L, by definition, has a trivial kernel.

Corollary 2.7

Combining the above observations, we find that L is an isomorphism $\Phi \to \mathbb{R}$ i.e. it respects the structure of either space s.t. its action is merely to relabel the concerned sets and their elements.

Corollary 2.8

L^{-1} is $\mathbb{R}$-linear.

Proof. As L is an isomorphism, so is L^{-1}. Therefore, L^{-1} is a linear map $\mathbb{R} \to \Phi$. □

Now, consider the commutative diagram below:

$$\begin{tikzcd} \mathbb{R} \arrow[r, bend left, "\varphi"] & \Phi \arrow[l, bend left, "L"] \end{tikzcd}$$

Figure 3.1: Some interesting maps

It turns out that given some L, we can define a ring homomorphism using the above diagram.

Theorem 2.9: A class of ring homomorphisms on $\mathbb{R}$

Invertible linear functionals (i.e. isomorphisms) $L : \Phi \to \mathbb{R}$ index ring homomorphisms $\varphi_L : \mathbb{R} \to Z(\Phi)$, where,

$$\varphi_L = L^{-1}$$
$$\varphi_L \{t\} = L^{-1} \{t\}$$

Proof. □

2.5 As producing a class of submanifolds of $\mathbb{R}^2$

A last piece of useful information (for now) about the structure of $(0, 1)$ is that

3 Conformal structure

The power of the above apparatus is that its structure remains essentially the same even if we upgrade Φ to a *noncommutative* division algebra over a *noncommutative ring* R endowed with *Zariski* topology! Let us see how.